

Alan M. Tommy

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SUMMARY

Computer science graduate with experience spanning full-stack development, data analytics, and artificial intelligence. Self-motivated and versatile, proven to deliver under pressure across diverse technical domains and team environments.

EDUCATION

University of Michigan, Ann Arbor, MI

Bachelor of Science in Engineering, Computer Science – May 2025

GPA: 3.74/4.00

Key Courses

Statistics and Artificial Intelligence, Software Engineering, Operating Systems, Web Systems, Computer Security, Machine Learning, Data Analytics Tools and Techniques, Computer Game Design and Development

EXPERIENCE

University of Michigan ROTC – Wargame Digitalization Team Intern

June 2025 – January 2026

- Worked in a team to develop a first-of-its-kind digital interface for a strategic military simulation on the web.
- Wrote more than 25% of the entire codebase, contributing to PostgreSQL database, Django backend, Redis-backed WebSockets, user authentication/authorization, and several frontend React components.
- Designed the database schema following normalization principles (3NF) and wrote code to efficiently populate and update it as gameplay state changed in real time.
- Re-architected the WebSocket server using multiplexing to reduce each client's connection count to a single WebSocket, improving scalability and reducing server load.
- Application has been demonstrated to successfully support **over 100 concurrent users and is being used to train and evaluate future officers** from the Army, Navy, Marine Corps, Air Force, and Space Force.

PROJECTS

Final Fumes (Video Game)

February 2025 – April 2025

- Developed a fully-fledged video game in Unity on a 4-person team within an 8-week Agile development cycle.
- Devised a procedural generation algorithm inspired by graph-based algorithms to spawn terrain while ensuring logical road connectivity.
- Showcased to live players; received overwhelmingly positive feedback, placing **4th among 20+ teams**.

Multi-Threaded Network File Server

November 2024 – December 2024

- Implemented a multi-threaded, crash-consistent network file server in C++20/Boost.Threads, handling concurrent client requests over raw TCP sockets.
- Enforced disk write ordering to guarantee on-disk file system consistency across crashes.
- Used fine-grained per-block locking to allow concurrent reads/writes to different disk blocks while serializing conflicting writes to the same block and safely parsed untrusted client input over the network protocol.

University of Michigan Ross Business+Tech Datathon

February 2024

- Collaborated with a 6-person team to analyze large carbon emissions datasets using Python (pandas, NumPy).
- Designed an environmentally and economically sustainable product in accordance with findings.
- Effectively communicated the product proposal to a panel of judges, placing **1st among 40+ teams**.

SKILLS

Programming Languages: C++, C, Python, SQL, JavaScript, TypeScript, Visual Basic, C#, R, MATLAB, Lua

Tools and Software: Linux/Unix, GitHub, Docker, Blender, Vim, Unity, Figma, Jira

Machine Learning/Artificial Intelligence: Prompt engineering, data preprocessing, model tuning, TensorFlow, PyTorch